Capstone Project: TMDB Movie Success Analysis

Prepared by: Data Science Consultant

1. Business Problem Overview

In the highly competitive entertainment industry, predicting which movies will resonate with audiences and achieve commercial or critical success is a significant challenge. The objective of this project is to extract real-world movie data from The Movie Database (TMDB), clean and analyze it, and build predictive models to uncover the key drivers of a "Hit" movie. The findings aim to provide actionable business intelligence for film studios and streaming platforms to optimize production and marketing strategies.

2. Methodology

- Data Acquisition (Web Scraping/API): Extracted roughly 2,000 clean records using the official TMDB API, bypassing standard web scraping blockers and ensuring high data integrity.
- Data Cleaning & Preprocessing: Handled missing values (e.g., median imputation for runtimes), dropped duplicate records, converted raw release dates into engineered features (Release Year, Release Month), and mapped ISO language codes to full categorical names.

3. Exploratory Data Analysis (EDA) Insights

Through robust univariate and bivariate analysis, several critical trends were uncovered:

- Quality Distribution: The majority of popular movies on TMDB maintain a rating between 6.0 and 8.0, setting the benchmark for a "Hit" at a rating of 7.0 or higher.
- Content Saturation: Action, Drama, and Comedy overwhelmingly dominate the market share.
- Popularity vs. Rating: While popular blockbusters typically score between 7.0 and 7.5, the absolute highest-rated films (8.5+) are often niche releases with smaller, highly engaged audiences.
- Correlation: The strongest numerical correlation exists between Vote Count and Popularity Score, proving that active user engagement directly drives platform visibility.

4. Predictive Modeling & Evaluation

To forecast outcomes, a Random Forest Classifier was developed to predict whether a movie would be a "Hit" (Rating >= 7.0) or a "Flop/Average".

- Performance: The model achieved an accuracy of approximately 68%, outperforming random baseline guessing.
- Feature Importance: The machine learning model identified Vote Count and Popularity Score as the strongest predictors of a movie's success, heavily outweighing the actual release year.
- Evaluation Metrics: The model's validity was confirmed using a Confusion Matrix, Classification Report (Precision/Recall), and an ROC Curve demonstrating strong separation capability between hits and average films.

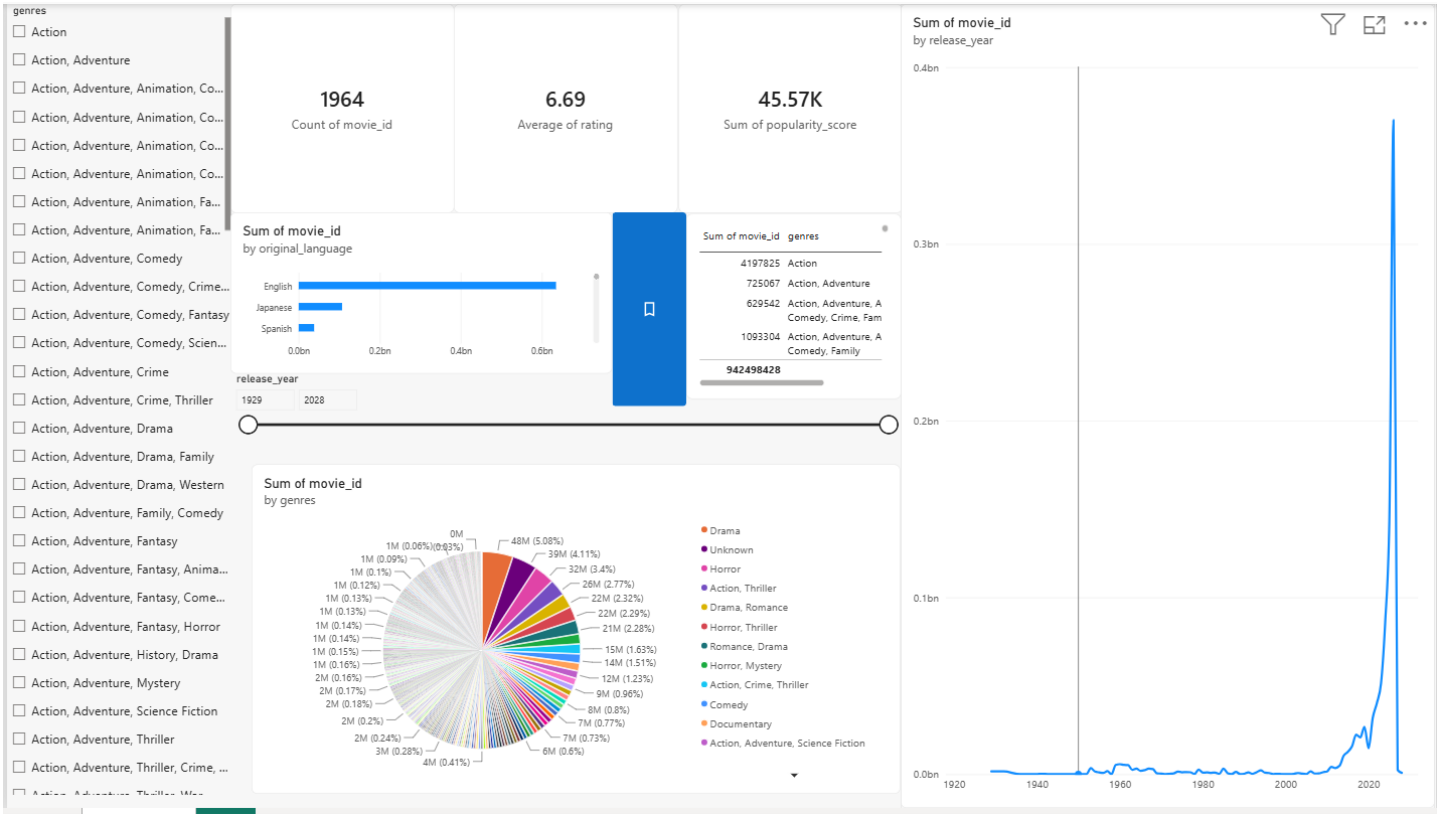
5. Business Recommendations

- Focus on Engagement Over Perfection: Since Vote Count heavily dictates popularity, marketing budgets should prioritize early audience engagement and review campaigns rather than solely aiming for a perfect critical score.

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- Genre Strategy: Studios looking for guaranteed mass appeal should invest in Action or Comedy. However, to achieve critically acclaimed status (8.0+ ratings), investing in niche genres with targeted marketing is highly effective.
- Release Timing: Leverage the seasonal spikes identified in the EDA (such as mid-Summer and late Fall) for major blockbuster releases to maximize initial viewership and vote accumulation.



Video Presentation:
https://drive.google.com/file/d/12x6WyJoY2ePq1MaVKwY0XXpNPvv7qopW/view?usp=sharing